

Kiran Raj

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SUMMARY

Python backend engineer with 3 years of experience building machine learning powered REST APIs. Deep expertise in NLP using BERT and sentence-transformers for semantic similarity, text classification, and embedding generation. Proficient in Flask and FastAPI for API development, MongoDB for document storage, scikit-learn for ML pipelines, and PyMuPDF for PDF parsing and text extraction. Strong experience with Docker containerisation, Git version control, pytest unit testing, and agile development practices. Comfortable building full stack applications integrating React frontends with Python backends via REST APIs and CORS configuration.

EXPERIENCE

Machine Learning Engineer — NLP Systems, Bangalore

August 2022 – Present

- Built NLP pipelines using BERT and sentence-transformers to generate semantic embeddings for document similarity tasks, achieving cosine similarity scores with 94% accuracy against human evaluations.
- Developed REST APIs in Flask serving ML model inference with sub-200ms response times, processing 100k+ daily requests.
- Used scikit-learn for cosine similarity computation, text vectorisation, and ML pipeline orchestration across multiple NLP projects.
- Integrated PyMuPDF for in-memory PDF text extraction in a document processing pipeline, handling multi-column and multi-page layouts.
- Designed MongoDB schemas for storing unstructured NLP results, embeddings metadata, and processing history with pymongo.
- Containerised all services using Docker and managed deployments with Docker Compose in a CI/CD pipeline via GitHub Actions.
- Wrote comprehensive unit tests using pytest achieving 90%+ coverage across all service modules.
- Collaborated in agile sprints using JIRA, participated in daily standups, sprint planning, and retrospectives.

Backend Engineer — DataLabs, Hyderabad

January 2021 – July 2022

- Built and maintained Python Flask REST APIs for a data analytics platform used by 50+ enterprise clients.
- Integrated BERT-based text classification models into production APIs for automated document tagging.
- Used MongoDB as primary datastore for storing user activity, API logs, and processed document metadata.
- Implemented CORS configuration to enable React frontend applications to communicate with Flask backend services.
- Worked with pandas and numpy for data preprocessing and feature engineering in ML workflows.

PROJECTS

AI Resume Matcher

Full stack resume screening tool using React, Flask, BERT sentence-transformers, MongoDB, and PyMuPDF. Encodes resume and job description as 384-dimensional BERT embeddings, computes cosine similarity for match scoring, performs keyword gap analysis using set subtraction, stores results in MongoDB, and displays animated SVG score ring in React.

Semantic Document Search Engine

Built a semantic search engine using sentence-transformers BERT model and FAISS for vector similarity search. Flask REST API backend, React search interface, MongoDB for document storage. Supports PDF upload with PyMuPDF text extraction and returns semantically similar documents using cosine similarity.

SKILLS

Languages: Python, JavaScript, SQL

Frameworks: Flask, FastAPI, React, scikit-learn, sentence-transformers, pandas, numpy

NLP / ML: BERT, cosine similarity, text classification, semantic embeddings, spaCy, FAISS

PDF: PyMuPDF (fitz), pdfminer, in-memory PDF text extraction

Databases: MongoDB (pymongo), PostgreSQL, Redis

DevOps: Docker, Docker Compose, GitHub Actions, Git

Testing: pytest, unit testing, integration testing

Practices: REST API design, CORS, agile, code review, CI/CD

EDUCATION

B.E. Computer Science — RV College of Engineering, Bangalore

2017 – 2021 | CGPA: 9.1 / 10